2 - Implement RK4

Use baseline timestep $\Delta t = 0.01$ and $T = 50$.

a. Run code with $\Delta t, \Delta t/2, \Delta t/4$.

b. Plot $x(t)$ for each run on the same graph. Comment on convergence with decreasing Δt.

c. Use built-in ODE solver and compare results.

Solution.

The following code was used to implement RK4 and run comparison with SciPy's built-in ODE solver, `solve_ivp` (similar to MATLAB's `ode45`):

Listing 1: Question 2 Python

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp

def F(u, sigma, rho, beta):
    x, y, z = u
    dxdt = sigma * (y - x)
    dydt = x * (rho - z) - y
    dzdt = x * y - beta * z
    return np.array([dxdt, dydt, dzdt])

def RK4_step(u, h, sigma, rho, beta):
    k1 = F(u, sigma, rho, beta)
    k2 = F(u + (h / 2) * k1, sigma, rho, beta)
    k3 = F(u + (h / 2) * k2, sigma, rho, beta)
    k4 = F(u + h * k3, sigma, rho, beta)

    y_new = u + (h / 6) * (k1 + 2 * k2 + 2 * k3 + k4)
    return y_new

def RK4_solver(u_0, h, sigma, rho, beta, T=50):
    num_steps = int(T / h)
    u_values = np.zeros((num_steps + 1, len(u_0)))
    u_values[0] = u_0

    for n in range(num_steps):
        u_values[n + 1] = RK4_step(u_values[n], h, sigma, rho, beta)

    return u_values
```

```python

if __name__ == "__main__":
    # < Some Styling Setup and Enable Latex >
    # ...
    # Use default parameters for Lorenz system (Case C)
    sigma, rho, beta = 10, 28, 8/3
    u_0 = np.array([1, 1, 1])
    dt = 0.01
    time_steps = [dt, dt/2, dt/4]

    # Run RK4 solver for each time step and plot results
    for h in time_steps:
        u_values = RK4_solver(u_0, h, sigma, rho, beta)
        time_values = np.arange(0, 50 + h, h)
        plt.plot(time_values, u_values[:, 0], label=rf"$\Delta t = {h}$")
    plt.xlabel(r"$t$")
    plt.ylabel(r"$x(t)$")
    plt.title("RK4 Results")
    plt.legend()
    plt.savefig('./outputs/rk4_q2.png', dpi=300)
    plt.cla()

    # Compare with SciPy's ODE solver
    scipy_solution = solve_ivp(
        lambda t, u: F(u, sigma, rho, beta),
        [0, 50],
        u_0,
        t_eval=np.arange(0, 50 + dt, dt)
    )

    # Plot Scipy solution
    plt.plot(scipy_solution.t, scipy_solution.y[0], label="Scipy", color=
   default_colors[3])
    plt.xlabel(r"$t$")
    plt.ylabel(r"$x(t)$")
    plt.title("Scipy Solver Results")
    plt.legend()
    plt.savefig('./outputs/rk4_scipy.png', dpi=300)
    plt.cla()
```

The following plots were obtained:

Figure 1: RK4 results for different time steps.

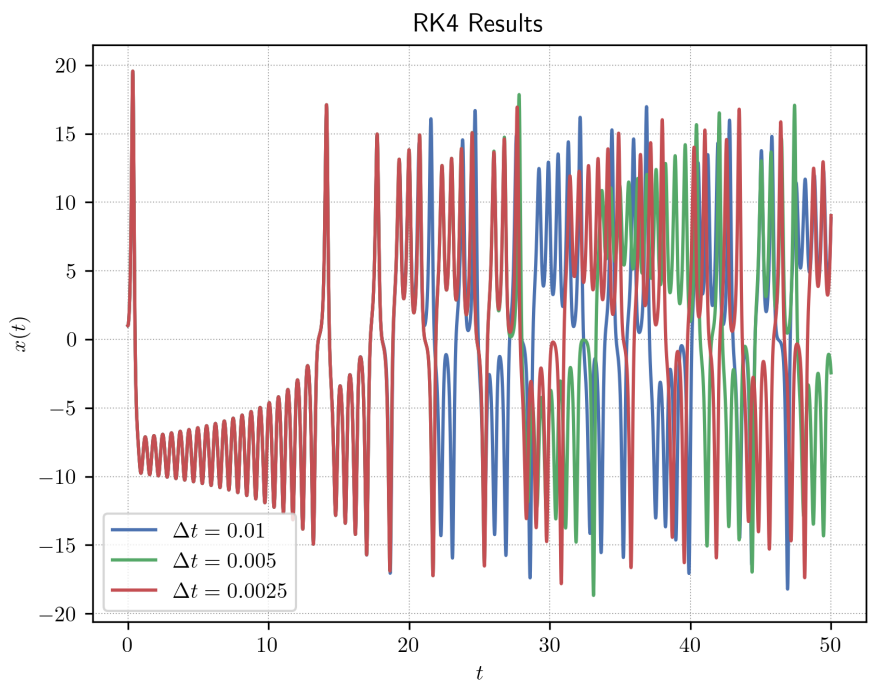
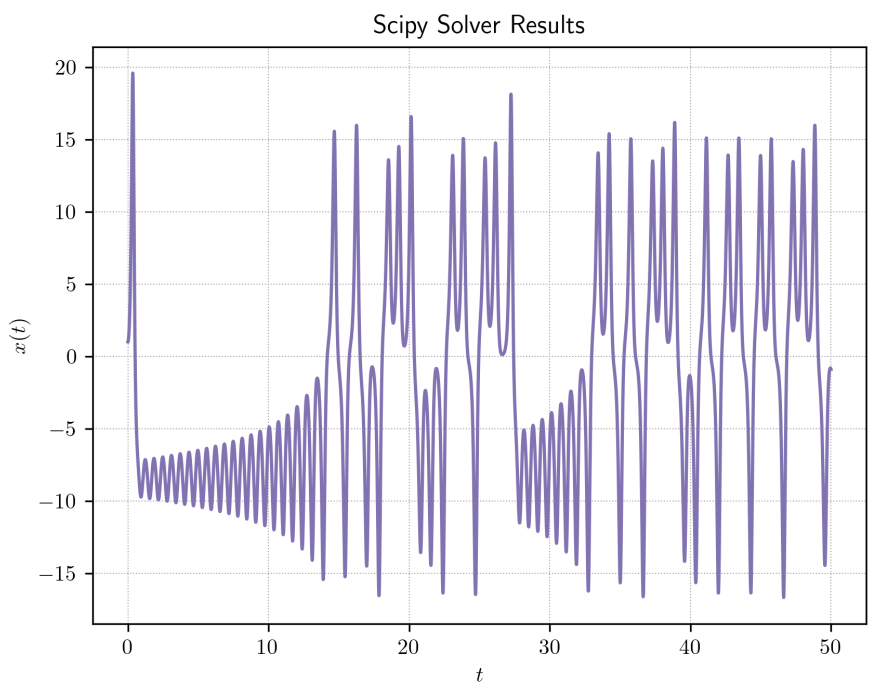


Figure 2: SciPy's `solve_ivp` results.



From Figure 1, we observe that as the time step Δt decreases, the solution seems to converge to a more consistent trajectory for larger time intervals. Or in other words, we see that the overlap for trajectories with smaller times steps is better before the Lorenz system's chaotic nature causes divergence.

In comparing Figures 1 and 2, we can see that the RK4 implementation closely matches the results from SciPy's `solve_ivp` up until about $t \approx 13$ (when graphs are overlapped). After this point, the trajectories begin to diverge due to numerical differences. This behavior is expected as `solve_ivp` uses an adaptive time-stepping method in comparison to our fixed time step RK4 implementation, which can lead to different numerical errors accumulating over time. However, since we observed that the initial trajectories match closely, we can conclude that our RK4 implementation is correct and captures the dynamics of the Lorenz system accurately for smaller time intervals.

---

3 - Parameter Exploration

Simulate $[0, T = 50]$ for the following parameter sets:

Case A:$(\sigma, \rho, \beta) = (10, 5, 8/3)$ **Case B:**$(\sigma, \rho, \beta) = (10, 15, 2)$

Case C:$(\sigma, \rho, \beta) = (10, 28, 8/3)$ **Case D:**$(\sigma, \rho, \beta) = (10, 40, 8/3)$

For each case:

a. Plot $x(t)$, $y(t)$, and $z(t)$ vs. t over $[0, T]$.

b. Plot the trajectory $(x(t), y(t), z(t))$ in phase space.

c. Comment on the behaviors of the system for each case.

Solution. The following code was used to implement RK4 and generate the required plots for each parameter set:

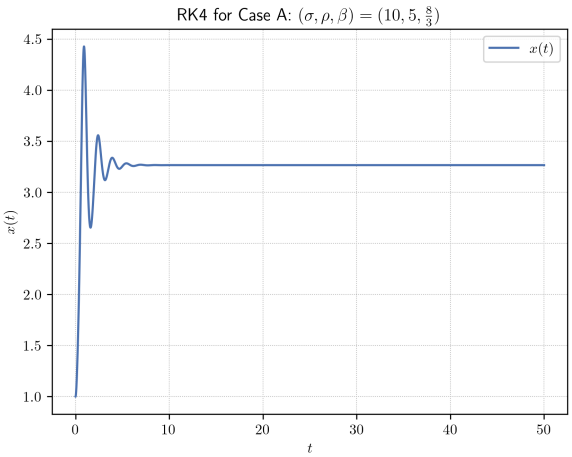
Listing 2: Question 3 Python

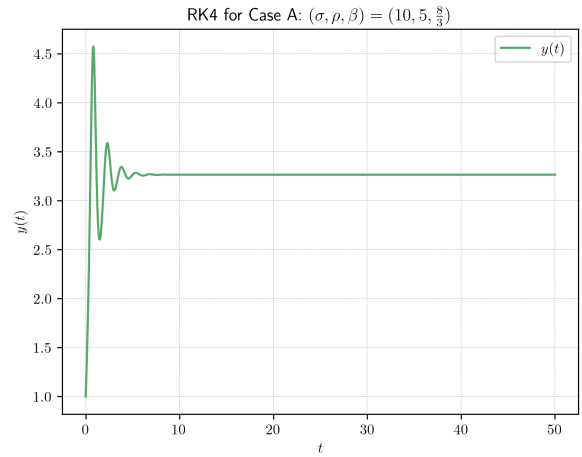
```python
if __name__ == "__main__":
    # < Some Styling Setup and Enable Latex >
    # ...
    # Get list of default colors for style
    prop_cycle = plt.rcParams["axes.prop_cycle"]
    default_colors = prop_cycle.by_key()["color"]

    # each element of param_sets is [sigma, rho, beta]
    param_sets = np.array(
        [
```

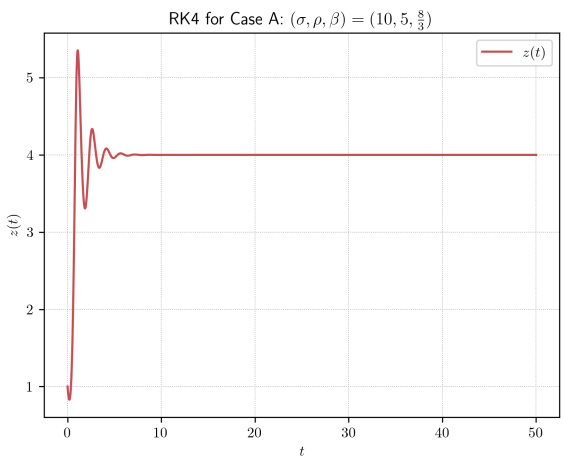
```
            # Case A
            [10, 5, 8 / 3],
            # Case B
            [10, 15, 8 / 3],
            # Case C
            [10, 28, 8 / 3],
            # Case D
            [10, 40, 8 / 3],
        ]
    )

    set_labels = ["A", "B", "C", "D"]

    u_0 = np.array([1, 1, 1])
    h = 0.01 # baseline time step
    time_values = np.arange(0, 50 + h, h)
    dimensions = ["x", "y", "z"]

    # Generate plots for each case
    for params, label in zip(param_sets, set_labels):
        u_values = RK4_solver(u_0, h, *params)

        # Plot x(t), y(t), z(t) individually
        for i, dim in enumerate(dimensions):
            fig, ax = plt.subplots()
            ax.plot(time_values, u_values[:, i], label=f"${dim}(t)$", color=default_colors[i])
            ax.set_xlabel(r"$t$")
            ax.set_ylabel(rf"${dim}(t)$")
            ax.set_title(rf'RK4 for Case {label}: $(\sigma, \rho, \beta) = (10, {int(params[1])}, \frac{8}{3})$')
            ax.legend()
            fig.savefig(rf"./outputs/rk4_{label}_{dim}.png", dpi=300)
            plt.close(fig)

        # Plot Phase Portrait
        fig = plt.figure()
        ax = fig.add_subplot(111, projection='3d')
        ax.plot(u_values[:, 0], u_values[:, 1], u_values[:, 2], color=default_colors[3])
        ax.set_xlabel(r'$x$')
        ax.set_ylabel(r'$y$')
        ax.set_zlabel(r'$z$')
        plt.title(rf'Phase Portrait for Case {label}: $(\sigma, \rho, \beta) = (10, {int(params[1])}, \frac{8}{3})$')
        plt.savefig(rf"./outputs/rk4_{label}_phase_portrait.png", dpi=300)
        plt.close(fig)
```

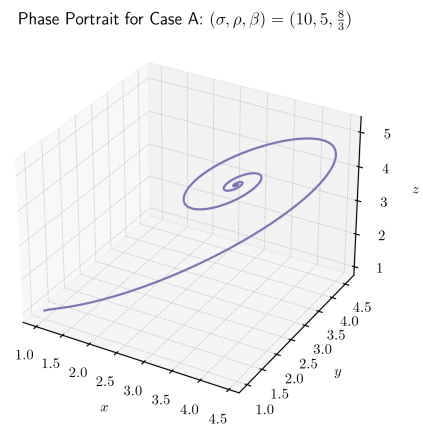
The following plots were obtained for each case:

Case A: $(\sigma, \rho, \beta) = (10, 5, 8/3)$



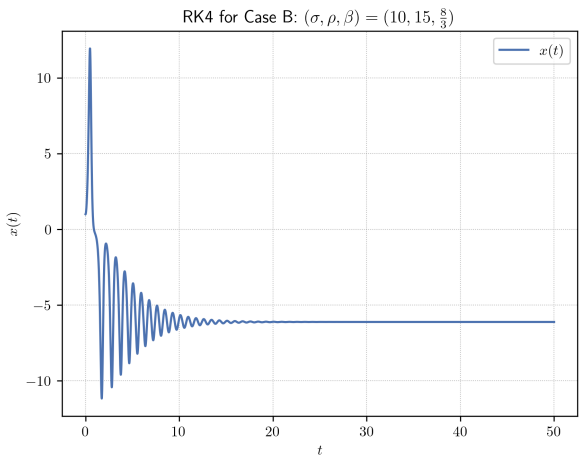


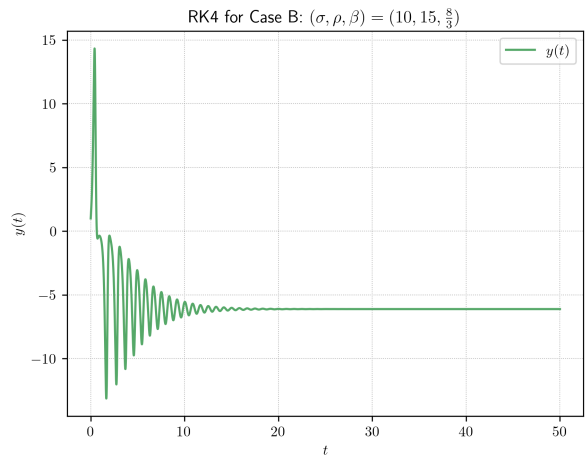


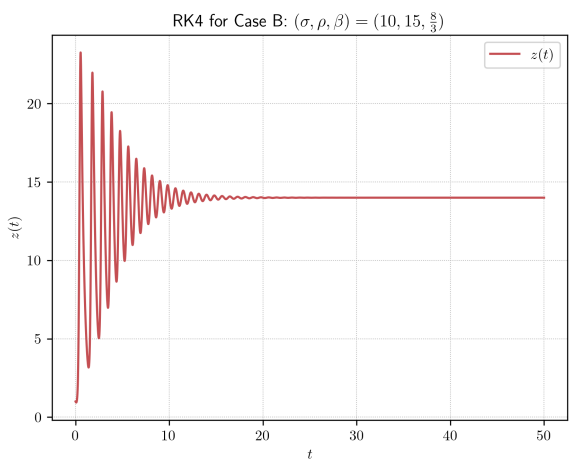


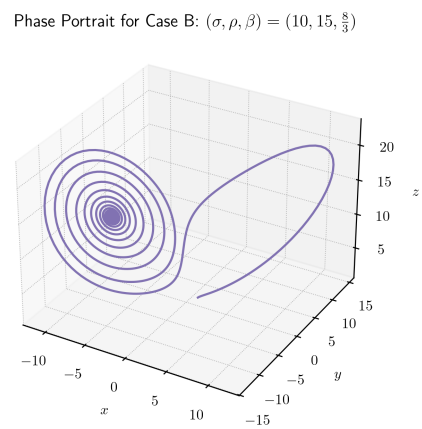
Comments: For Case A, we observe that the curve in the phase portrait converges to a spiral sink around $(3.25, 3.25, 4)$. The spiral itself quickly converges to the fixed point and oscillations in all directions are dampened.

Case B: $(\sigma, \rho, \beta) = (10, 15, 8/3)$



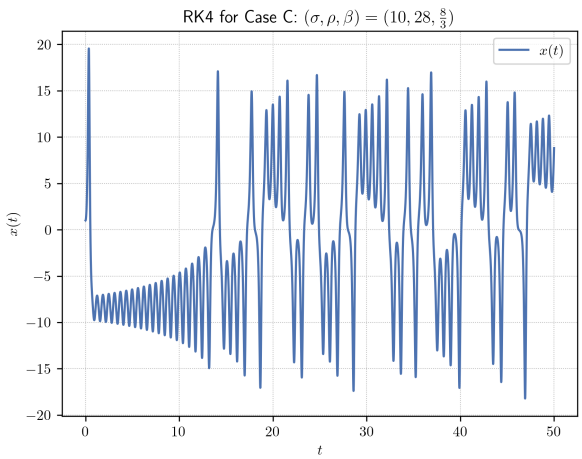


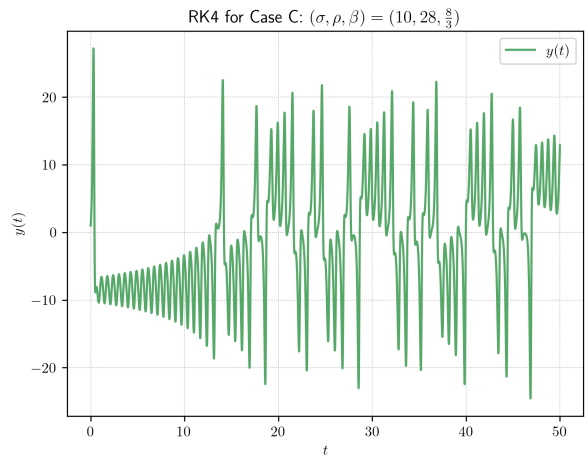


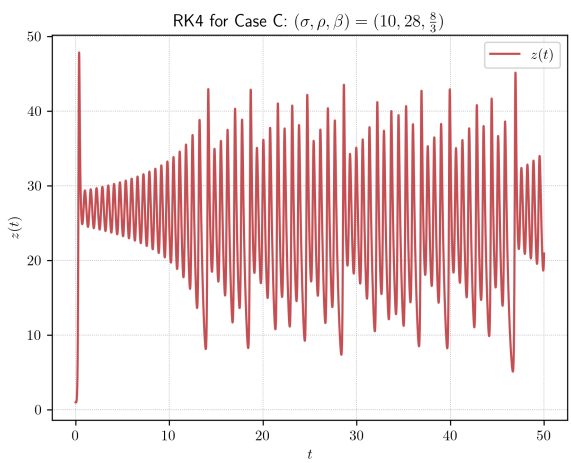


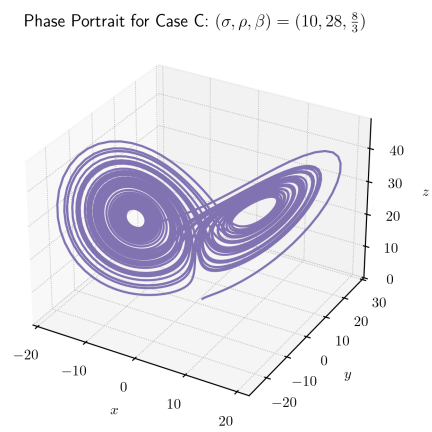
Comments: For Case B, we observe that the curve in the phase portrait converges to a spiral sink around $(-6, -6, 14)$. The increase in ρ caused the sinks to shift further from the origin and the oscillations to increase in amplitude. However, in this case, the system still converges to a fixed point (though slightly more slowly).

Case C: $(\sigma, \rho, \beta) = (10, 28, 8/3)$



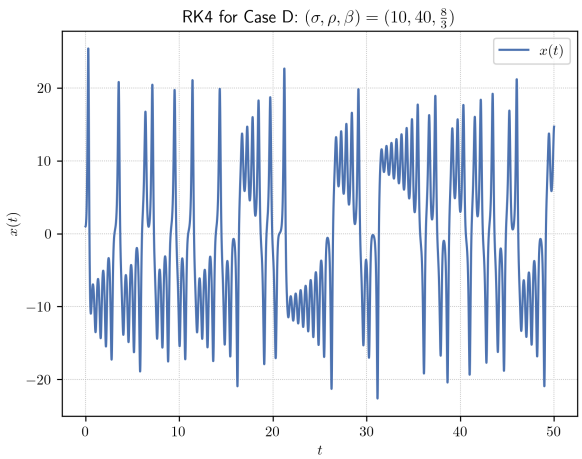


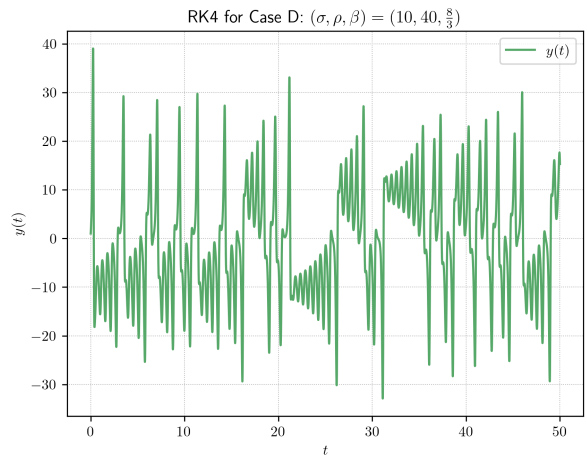


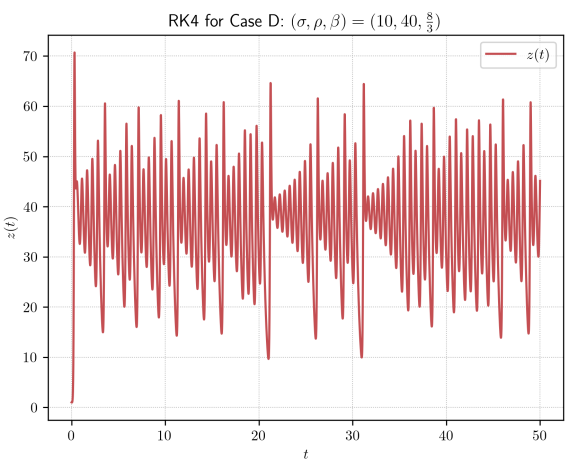


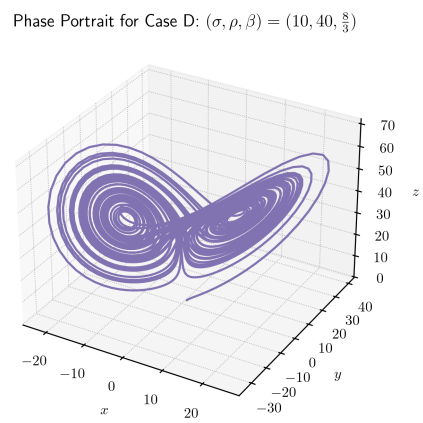
Comments: For Case C, we observe the classical Lorenz attractor in the phase portrait. The curve travels in a figure eight pattern around the two distinct sources. The trajectory does not converge to a fixed point. Instead the curve oscillates between the two lobes of the attractor indefinitely. With this increase in ρ, the system underwent a bifurcation, transitioning from fixed point convergence to looping behavior.

Case D: $(\sigma, \rho, \beta) = (10, 40, 8/3)$









Comments: For Case D, we observe that again the lobes shift further from the origin with the increase in ρ. The trajectory still oscillates between the two lobes of the attractor indefinitely. However, the looping has larger amplitudes and the trajectory itself is more erratic.

In general, as ρ increases, the system transitions from stable equilibrium behavior to more complex and chaotic dynamics. When ρ is sufficiently large, the sinks lose stability and the system exhibits the characteristic butterfly-shaped attractor of the Lorenz system. Where the trajectory oscillates between the two lobes of the attractor with no clear periodicity. Further increases in ρ lead to more erratic switching between the lobes and enlarges the attractor itself.

4 - Sensitivity Experiment (repeatability test)

Choose one of the four parameter sets from Question 3. Run two simulations with the following initial conditions:

$$(x_0, y_0, z_0) = (1, 1, 1) \quad (x_0, y_0, z_0) = (1 + 10^{-8}, 1, 1)$$

using the same Δt.

a. Plot $x(t)$ from both runs on the same graph. Comment on the results.

b. Plot the difference $|x_1(t) - x_2(t)|$ vs. t on a semilog-y plot (i.e., only the y-axis is logarithmic). Comment on the results.

c. Do the Same for $y(t)$ and $z(t)$.

Solution. temp text

5 - Time Step Effects and Numerical Stability

For Case D, repeat the computation with:

$$\Delta t = 0.02, 0.01, 0.005$$

a. Compare the time series plots and the phase-space trajectories.

b. Describe any visible artifacts (eg. drift, distortion of trajectories, blow-up)

c. Describe the influence of step size from the simulation results.

Solution. dummy text